FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 1HA-I 24006 — 30062 (Version B) / 32062 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

BIOLOGY HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 30062 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Which protein provides support for tendons and ligaments?

(A) Histone
(B) Elastin
(C) Casein
(D) Keratin

Q2. RNA lacks:

(A) Cytosine
(B) Adenine
(C) Thymine
(D) Guanine

Q3. High specific heat capacity of water helps in:

(A) Thermo stabilization
(B) Producing cooling effect
(C) Maintaining pH of medium
(D) Transport of materials

Q4. Absorption of different wavelengths by photosynthetic pigments is measured by:

(A) Differential staining
(B) Chromatography
(C) Spectrophotometry
(D) Electrophoresis

Q5. Which one of the following enzymes is NOT present in core of HIV?

(A) Protease
(B) Reverse transcriptase
(C) Integrase
(D) Transferase

Q6. In which phase of bacterial growth, 'only increase in cell size' occurs?

(A) Decline phase
(B) Lag phase
(C) Log phase
(D) Stationary phase

Q7. Which of the following is a bacterial disease?

(A) Sleeping sickness
(B) Herpes
(C) Malaria
(D) Ulcer

Q8. The photosynthetic partner in a lichen is often:

(A) Diatoms
(B) Green algae
(C) Brown algae
(D) Dino flagellates

Q9. The cell supporting herbaceous parts of plants having extra cellular deposition of cellulose and living protoplasm are:

(A) Parenchyma
(B) Sclerenchyma
(C) Collenchyma
(D) Tracheids

Q10. Liver is NOT involved in the synthesis of:

(A) Heparin
(B) Albumin
(C) Vitamin A
(D) Cholecystokinin

Q11. Identify the characteristic related to chondrichthyes:

(A) Placoid scales on body
(B) Homocercal caudal fin
(C) Presence of operculum
(D) Swim bladder for buoyancy

Q12. Wood produced by the activity of vascular cambium is also called:

(A) Inner bark
(B) Phellogen
(C) Secondary xylem
(D) Secondary phloem

Q13. In the given diagram, bundle of His are represented by: (A heart diagram with regions labelled I, II, III, IV is printed in the source scan.)

(A) IV
(B) I
(C) II
(D) III

Q14. Identify the phylum with bilateral symmetry, saclike digestive system and no coelom:

(A) Aschelminthes
(B) Cnidaria
(C) Nematoda
(D) Platyhelminthes

Q15. Dicots have:

(A) Reticulate venation in leaf
(B) Fibrous roots
(C) Floral parts multiple of 3
(D) Dispersed vascular bundles

Q16. Morels and truffles are examples of:

(A) Basidiomycota
(B) Zygomycota
(C) Deuteromycota
(D) Ascomycota

Q17. Natural rubber is a type of:

(A) Waxes
(B) Phospholipids
(C) Terpenes
(D) Steroids

BIOLOGY HSSC-I — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 24006-(B) (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 42) — Q.2 Answers the following questions briefly. (14 x 3 = 42)

درج سوالات کے مختصر جوابات تحریر کریں۔

(i) [2+1] Elaborate the role of peroxisomes in plant and animal cells. — OR (03) — Relate variety of proteins of plasma membrane with their respective roles. (Any three)

(ii) [03] Illustrate the formation of glycosidic bonds in: a Sucrose b Maltose — OR (03) — Illustrate the process of non-cyclic photophosphorylation with labelling.

(iii) [03] Justify the significance of amino acid sequence by explaining example of sickle cell haemoglobin. — OR (03) — Outline the process of C4 photosynthesis in a flow chart.

(iv) [03] How do oxidoreductases, hydrolases and lyases work? Also give one example of each group. — OR (2+1) — How does endospore formation occur in bacteria? How does it help bacteria to withstand unfavorable conditions?

(v) [03] Describe steps of lytic cycle of bacteriophage. — OR (03) — Explain pathogenic role of fungi in human with their symptoms. (Any three)

(vi) [03] A person was bitten by snake. Antivenom was injected. a How does antivenom work? b Why was passive immunity preferred? — OR (1x3) — A bacterium is shown in the diagram: a Name the structures P and R. b Write chemical composition of Q. c Name the process by which bacteria reproduces asexualy. (A bacterium diagram labelled P, Q, R is printed in the source scan.)

(vii) [03] T-cells are involved in specific defence. Name any three types of T-cells with their specific roles. — OR (03) — Write about evolutionary adaptations in phylum arthropoda regarding respiration, excretion and nervous system.

(viii) [03] Briefly explain the flow of blood through heart as regulated by valves. — OR (03) — Write down steps of swallowing action of bolus in oral cavity.

(ix) [2+1] a How does temperature affect rate of an enzyme catalyzed reaction? b Compare optimum temperature of enzymes in human and thermophilic bacteria in a graph. — OR (03) — Name any three groups of Protista with one salient feature and example of each group.

(x) [2+1] In TACT theory of ascent of sap, how transpiration and adhesion help in process? — OR (03) — Name and explain two hypotheses for evolution of single veined leaves.

(xi) [03] Complete the table: | Phylum | Distinguishing character | Example | | a | Cnidarian | | | | b | | Metameric segmentation | | | c | | | Snail | — OR (03) — Life cycle of fern is shown in diagram: a Name generations P and Q. b Which processes are shown by X and Y? c Which generation is dominant in ferns? (A fern life-cycle diagram labelled P, Q, X, Y is printed in the source scan.)

(xii) [03] Write causative agent of cotton leaf curl disease. Also write symptoms and treatment of disease. — OR (03) — Write down the mechanism of translocation in plants.

(xiii) [03] What are xerophytes? Enlist any four adaptations for their habitat. Also give example. — OR (03) — How is dermis of skin involved in first line of defence?

(xiv) [03] Name two growth promoter and one growth inhibitor in plants with one major effect of each on plant body. — OR (03) — Write mechanism of irreversible non-competitive enzyme inhibition with an example.

> Note: SECTION – C (Marks 26) — Attempt the following questions.

Q3. (Marks 06) Q.3 Spermatophytes are seeded plants. How did seeds evolve? — OR (3+3) — How are influx and efflux of potassium ions involved in regulating opening and closing of stomata? Explain in detail.

Q4. (Marks 2+5) Q.4 Explain mechanical and chemical digestion in human stomach. — OR (2+4+1) — Describe the structure and functions of Golgi complex. Also draw its diagram.

Q5. (Marks 06) Q.5 Give a detailed account of general characteristics of class Aves. — OR (3+3) — Explain steroids and prostaglandins as important group of lipids. Describe their roles in living organisms.

Q6. (Marks 5+2) Q.6 Bacteria show para sexuality. Explain transduction and transformation. Also draw diagrams showing processes. — OR (1+5+1) — What is an electro cardio gram? Explain its different components. Also draw a neat sketch.

VERSION D

BIOLOGY HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 32062 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Identify the process occurring in bacteria in given diagram: (A donor-cell to recipient-cell diagram is printed in the source scan.)

(A) Transformation
(B) Transduction
(C) Conjugation
(D) Translation

Q2. Glyoxisomes will most probably be abundant in the seeds of:

(A) Wheat
(B) Pea
(C) Beans
(D) Peanut

Q3. Which property of water helps to stabilize lipid bilayer of plasma membrane?

(A) Cohesion
(B) Ionization
(C) Hydrophobic exclusion
(D) High polarity

Q4. The disease caused by viroids in human is:

(A) Hepatitis D
(B) Hepatitis A
(C) Hepatitis B
(D) Hepatitis C

Q5. The difference between photosynthesis of cyanobacteria and algae is in:

(A) Release of oxygen
(B) Accessory pigments
(C) Source of carbon
(D) Source of hydrogen

Q6. Which one of the following is an alga?

(A) (A circular microorganism diagram is printed in the source scan.)

(B) (An elongated microorganism diagram is printed in the source scan.)

(C) (An irregular microorganism diagram is printed in the source scan.)

(D) (A mushroom diagram is printed in the source scan.)

Q7. All the following characteristics are related to monocots EXCEPT:

(A) Vascular bundles in a ring

(B) Floral parts multiple of three

(C) Fibrous roots

(D) Parallel veins

Q8. In phylum Platyhelminthes, most systems are poorly developed EXCEPT:

(A) Excretory system

(B) Reproductive system

(C) Digestive system

(D) Nervous system

Q9. Phellogen is also called:

(A) Vascular cambium

(B) Intercalary meristem

(C) Lateral meristem

(D) Cork cambium

Q10. Water potential Ψ_w, solute potential Ψ_s and pressure potential Ψ_p are interrelated. If in a cell Ψ_w = -2100 kPa, Ψ_s = -3000 kPa, then Ψ_p will be:

(A) 5100 kPa

(B) -5100 kPa

(C) -900 kPa

(D) 900 kPa

Q11. Increase in proportion of unsaturated fatty acids in plasma membrane is an adaptation in plants living in:

(A) Water logged environment

(B) Saline environment

(C) Cold environment

(D) Hot environment

Q12. The mechanical tissue in seed coat are:

(A) Collenchyma

(B) Fibers

(C) Sclerids

(D) Parenchyma

Q13. If bolus accidentally enters larynx instead of esophagus, it may be due to:

(A) Lubrication in oral cavity

(B) Wave of contraction and relaxation

(C) Downward movement of larynx

(D) Upward movement of palate

Q14. The thickest layer of heart is:

(A) Pericardium

(B) Epicardium

(C) Endocardium

(D) Myocardium

Q15. Identify the vein which does NOT open in vena cava?

(A) Pulmonary vein

(B) Hepatic vein

(C) Iliac vein

(D) Jugular vein

Q16. Which pair describes aortic valve? I. Semilunar valve II. Between left ventricle and aorta III. Atrio ventricular valve IV. Passes deoxygenated blood

(A) I, IV
(B) I, II
(C) II, III
(D) III, IV

Q17. The technique used to separate photosynthetic pigments is:

(A) Chromatography
(B) Electrophoresis
(C) Spectro-photometry
(D) Centrifugation

BIOLOGY HSSC-I — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 24006-(D) (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 42) — Q.2 Answers the following questions briefly. (14 x 3 = 42)

درج سوالات کے مختصر جوابات تحریر کریں۔

(i) [03] The diagram represents a neutral lipid: a Name the components X and Y. b Name type of bond between X and Y. c Why this molecule is called neutral lipid? (A neutral-lipid structure diagram labelled X and Y is printed in the source scan.) — OR (03) — Complete the table with reference to bacterial cell wall: | Character | Gram-ive | Gram+ive | | Porins | | | | Peptidoglycans | | | | Thickness | | |

(ii) [1+2] What are mesophytes? How they adapt to their environment? (Any four adaptions) — OR (03) — What are evolutionary adaptations in echinoderms regarding digestion, respiration and nervous system?

(iii) [03] Differentiate between primary and secondary growth. (Any three differences) — OR (03) — Differentiate between Mitochondria and Chloroplasts. (Any three differences)

(iv) [03] How does natural killer cell kill, 'cells infected by bacteria' and 'cancerous cells'? — OR (03) — Laboratory manufactured sugars are not metabolized by enzymes in body. Justify.

(v) [2+1] How does pH affect the rate of an enzyme action? Compare the optimum pH for trypsin and papain. — OR (03) — How does temperature affect activities of RuBisCO?

(vi) [03] Complete the table: | Character | Animal | Plant | Fungus | | Mode of Nutrition | | | | | Cell wall composition | | | | | Presence of centriole | | | | — OR (03) — A student accidentally got a small cut on finger in lab. What series of events would occur as inflammatory response to this injury?

(vii) [03] Draw the structure of an antibody. Label its parts. Write their specific roles. — OR (03) — Draw an outline of Calvin cycle mentioning substrates and products of each step.

(viii) [03] a Why the Human Immunodeficiency Virus is called so? b Name two opportunistic diseases caused by HIV. c Name any two enzymes present in HIV core. — OR (1+2) — a Why

bryophytes are called amphibious plants? b Write any two distinguishing features of bryophytes.

(ix) [03] How does development of protostomes and deuterostomes differ in cleavage, coelom formation and blastopore fate? — OR (03) — Draw and label the steps of lysogenic cycle of bacteriophage.

(x) [03] How does blood circulation occur between heart and kidneys? Elaborate the answer. — OR (03) — What is feedback inhibition in enzymes? Elaborate with a proper example.

(xi) [1+2] Enlist the parts of large intestine. Also write the specific roles of large intestine. — OR (03) — What is photoperiodism? How does it affect short day and long day plants? Give one example of each.

(xii) [03] Briefly explain Racemose and Cymose types of inflorescence. — OR (1+2) — What is Mycorrhizae? Name and differentiate between its two types.

(xiii) [03] Name the parts of a bacterial flagellum. Also write their structures. — OR (03) — Briefly describe any three benefits of bacterial flora to humans.

(xiv) [03] Complete the table: | Group | Character | Example | | 1 | Whisk fern | | | | 2 | | Jointed stem | | | 3 | | | Adiantum | — OR (03) — Gametophyte of ferns is shown in the diagram: a What is other name for this structure? b Label parts A and B. c How does gametophyte of ferns differ from other plants' gametophytes? (A fern gametophyte diagram labelled A and B is printed in the source scan.)

> Note: SECTION – C (Marks 26) — Attempt the following questions.

Q3. (Marks 4+2) Q.3 Explain the components and mechanism of electron transport chain in mitochondria. Also draw the flow sheet. — OR (06) — Describe general characteristics of class Mammalia.

Q4. (Marks 3+2+2) Q.4 Describe the role of Lysosomes in Autophagy, intracellular digestion and autolysis. (Diagram is not required) — OR (4+3) — Explain the structure and role of different components of conducting system of human heart.

Q5. (Marks 2+2+2) Q.5 Write down general characteristics of polysaccharides. Describe characteristics and molecular structure of starch and cellulose. — OR (06) — Explain the mechanisms of pathways taken by water to reach xylem tissue in plants.

Q6. (Marks 2+3+2) Q.6 Explain development of male and female gametophytes in flowerings plants. Also draw life cycle of a flowering plant. — OR (2+5) — Outline the structure of pancreas and explain its role as an exocrine gland.